

PDF PASSWORD RECOVERY & HASH CRACKING REPORT

OFFLINE (JOHN THE RIPPER) & ONLINE (NETWORKWALKS) METHODS

W3-PM1 & W3-PM2 | CYBERSECURITY | NETWORKWALKS

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Program / Batch	B083F - Networkwalks
Date	26 September 2026
Modules Completed	W3-PM1 (Kali Linux & John the Ripper) W3-PM2 (Networkwalks Online Hash & Password Cracking Suite)
Target Files	My Locked PDF1.pdf, My Locked PDF2.pdf, My Locked PDF3.pdf (own lab-provided password-protected PDFs)
Permission secured from client?	Yes - self-owned lab files supplied by Networkwalks for internship practice
Phases Covered	Phase 1: Offline Hash Extraction & Dictionary Attack (JTR) Phase 2: Online Hash Extraction & Dictionary Attack (Networkwalks Suite) Phase 3: Verification & Flag Capture

1. Liability Disclaimer

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2. Introduction

This report covers Week 3 of my ongoing cybersecurity internship at Networkwalks, focused on password recovery and hash cracking of encrypted PDF documents. The task was performed using two complementary approaches: an offline attack inside Kali Linux using John the Ripper and pdf2john.pl (W3-PM1), and an online browser-based attack using the Networkwalks Hash Calculator and Password Cracker tools (W3-PM2).

Three password-protected PDF files - My Locked PDF1.pdf, My Locked PDF2.pdf and My Locked PDF3.pdf - were used as the target files. For each file, a crackable \$pdf\$ hash was first extracted, then a dictionary attack was run against that hash to recover the plain-text password, and finally the PDF was opened with the recovered password to capture the embedded verification flag.

Every step below includes the exact command or action used, the result observed, a screenshot as evidence, and a short note on why the finding matters from a security perspective.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used for offline hash extraction and dictionary attacks
John the Ripper (JTR)	Offline password-cracking tool used to brute force / dictionary-attack the PDF hashes
pdf2john.pl	Perl utility (bundled with JTR) used to extract a crackable \$pdf\$ hash from an encrypted PDF
rockyou.txt	Standard password wordlist used as the dictionary source for the attack
Networkwalks Hash Calculator	Online browser-based tool used to extract the \$pdf\$ hash from an encrypted PDF without Kali
Networkwalks Password Cracker	Online dictionary-attack tool used to crack the extracted \$pdf\$ hash and reveal the password
PDF Reader (qpdfview / browser PDF viewer)	Used to open the unlocked PDF with the recovered password and capture the verification flag

4. Activities Performed

4.1 Offline Password Cracking with John the Ripper (W3-PM1)

I performed offline hash extraction and dictionary attacks against all three target PDFs inside a Kali Linux virtual machine using pdf2john.pl and John the Ripper with the rockyou.txt wordlist.

Step 1 – Environment Verification & Directory Setup

I checked the installed John the Ripper version and located the pdf2john.pl helper script, then confirmed the target PDFs (Profile files and the three ‘My Locked PDF’ files) were present in the ~/W3-PM1 working directory.

```
MT Linux (Week1-Lab-Configured) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali -
Session Actions Edit View Help
-rwxr-x-- 1 kali kali 39K Sep 26 06:43 'Profile (1).pdf'
-rwxr-x-- 1 kali kali 42K Sep 26 06:43 'Profile (2).pdf'
-rwxr-x-- 1 kali kali 42K Sep 26 06:43 'Profile (3).pdf'
-rwxr-x-- 1 kali kali 832K Sep 26 06:43 'SIH26105-X-Arise - Copy.pdf'

(kali@kali)-[~/W3-PM1]
$ cd ..

(kali@kali)-[~]
$ john --version
Unknown option: "--version"

(kali@kali)-[~]
$ find /usr/share/john -name "pdf2john.pl" 2>/dev/null
/usr/share/john/pdf2john.pl

(kali@kali)-[~]
$
```

Screenshot 1: Verifying pdf2john.pl location and listing target PDF files

```
MT Linux (Week1-Lab-Configured) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali -~/W3-PM1
Session Actions Edit View Help
(kali@kali)-[~/W3-PM1]
$ ls -lh ~/W3-PM1
total 69M
-rwxr-x-- 1 kali kali 3.5M Sep 26 06:43 '03, SEPTEMBER, 2026.pdf'
-rwxr-x-- 1 kali kali 3.5M Sep 26 06:43 '1st slide.pdf'
-rwxr-x-- 1 kali kali 1.1M Sep 26 06:43 'Bharani (1).pdf'
-rwxr-x-- 1 kali kali 3.4M Sep 26 06:43 'Cyber Class.pdf'
-rwxr-x-- 1 kali kali 554K Sep 26 06:43 'first dample.pdf'
-rwxr-x-- 1 kali kali 27M Sep 26 06:43 'GUNASEKARAN (1)_merged.pdf'
-rwxr-x-- 1 kali kali 24M Sep 26 06:43 'GUNASEKARAN (1).pdf'
-rwxr-x-- 1 kali kali 4.9M Sep 26 06:43 'Intellect Hack 2026 - PPT.pdf'
-rwxr-x-- 1 kali kali 169K Sep 26 06:43 'intern report.pdf'
-rwxr-x-- 1 kali kali 267K Sep 26 06:56 'My Locked PDF1.pdf'
-rwxr-x-- 1 kali kali 205K Sep 26 06:56 'My Locked PDF2.pdf'
-rwxr-x-- 1 kali kali 314K Sep 26 06:56 'My Locked PDF3.pdf'
-rwxr-x-- 1 kali kali 39K Sep 26 06:43 'Profile (1).pdf'
-rwxr-x-- 1 kali kali 42K Sep 26 06:43 'Profile (2).pdf'
```

Screenshot 2: Confirming My Locked PDF1/2/3.pdf inside ~/W3-PM1

Step 2 – Wordlist Preparation & Hash Extraction

I extracted the rockyou.txt wordlist and then ran pdf2john.pl against My Locked PDF1.pdf to generate a crackable \$pdf\$ hash, saving it to hashes/pdf1.hash. The extracted hash showed the encryption as Revision 4 / Version 4 / 128-bit RC4/AES.

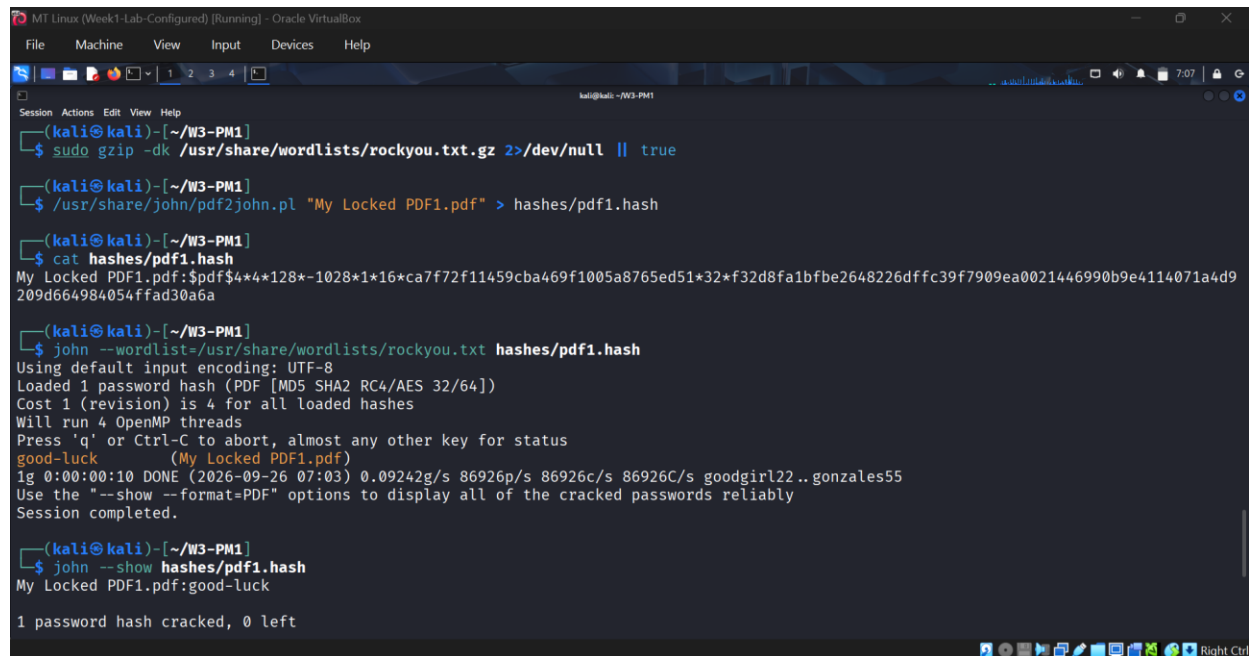
```
(kali㉿kali)-[~/W3-PM1]
$ cat hashes/pdf1.hash
My Locked PDF1.pdf:$pdf$4*4*128*-1028*1*16*ca7f72f11459cba469f1005a8765ed51*32*f32d8fa
209d664984054ffad30a6a
```

Screenshot 3: \$pdf\$ hash extracted from My Locked PDF1.pdf using pdf2john.pl

Step 3 – Dictionary Attack & Password Recovery (PDF1, PDF2, PDF3)

I repeated the pdf2john.pl extraction for all three files and ran john --wordlist=/usr/share/wordlists/rockyou.txt against each hash file, then used john --show to display the recovered credential.

Result – My Locked PDF1.pdf: cracked in 10 seconds. Recovered password: good-luck.



```
MT Linux (Week1-Lab-Configured) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/W3-PM1
Session Actions Edit View Help
(kali㉿kali)-[~/W3-PM1]
$ sudo gzip -dk /usr/share/wordlists/rockyou.txt.gz 2>/dev/null || true

(kali㉿kali)-[~/W3-PM1]
$ /usr/share/john/pdf2john.pl "My Locked PDF1.pdf" > hashes/pdf1.hash

(kali㉿kali)-[~/W3-PM1]
$ cat hashes/pdf1.hash
My Locked PDF1.pdf:$pdf$4*4*128*-1028*1*16*ca7f72f11459cba469f1005a8765ed51*32*f32d8fa1bfbe2648226dfc39f7909ea0021446990b9e4114071a4d9
209d664984054ffad30a6a

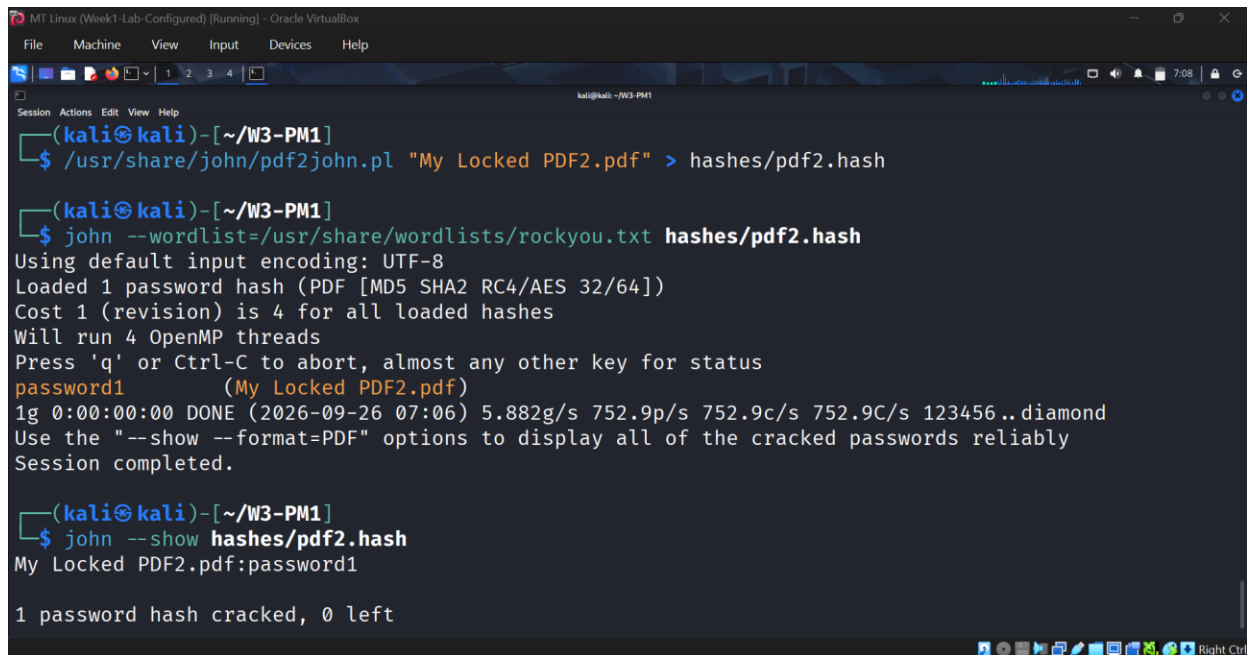
(kali㉿kali)-[~/W3-PM1]
$ john --wordlist=/usr/share/wordlists/rockyou.txt hashes/pdf1.hash
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
good-luck (My Locked PDF1.pdf)
1g 0:00:00:10 DONE (2026-09-26 07:03) 0.09242g/s 86926p/s 86926c/s 86926C/s goodgirl22..gonzales55
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.

(kali㉿kali)-[~/W3-PM1]
$ john --show hashes/pdf1.hash
My Locked PDF1.pdf:good-luck

1 password hash cracked, 0 left
```

Screenshot 4: John the Ripper cracks My Locked PDF1.pdf – password ‘good-luck’

Result – My Locked PDF2.pdf: cracked instantly. Recovered password: password1.

A screenshot of a terminal window titled "MT Linux (Week1-Lab-Configured) [Running] - Oracle VirtualBox". The terminal shows a user at the prompt "(kali㉿kali)-[~/W3-PM1]" running the command "\$ /usr/share/john/pdf2john.pl 'My Locked PDF2.pdf' > hashes/pdf2.hash". Then, they run "\$ john --wordlist=/usr/share/wordlists/rockyou.txt hashes/pdf2.hash". The output shows it loaded 1 password hash, cost 1, and ran 4 OpenMP threads. It successfully cracks the password "password1" with a speed of 5.882g/s. Finally, they run "\$ john --show hashes/pdf2.hash" which displays "My Locked PDF2.pdf:password1" and reports "1 password hash cracked, 0 left".

```
(kali㉿kali)-[~/W3-PM1]
$ /usr/share/john/pdf2john.pl "My Locked PDF2.pdf" > hashes/pdf2.hash

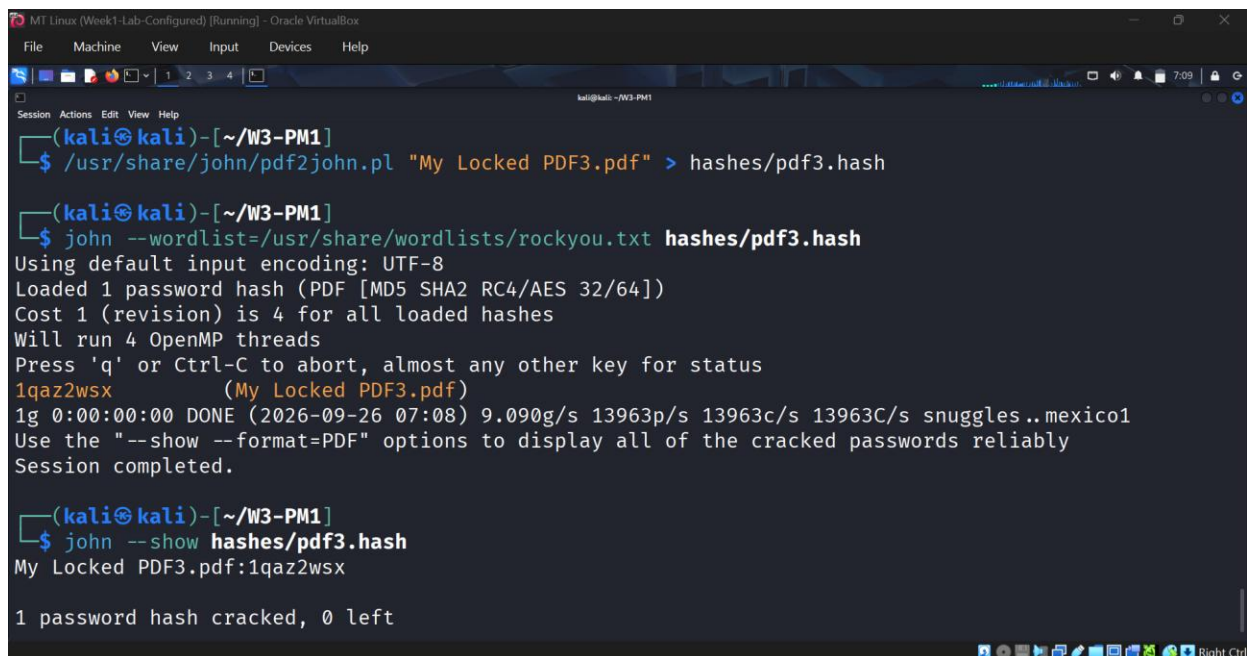
(kali㉿kali)-[~/W3-PM1]
$ john --wordlist=/usr/share/wordlists/rockyou.txt hashes/pdf2.hash
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
password1 (My Locked PDF2.pdf)
1g 0:00:00:00 DONE (2026-09-26 07:06) 5.882g/s 752.9p/s 752.9c/s 752.9C/s 123456..diamond
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.

(kali㉿kali)-[~/W3-PM1]
$ john --show hashes/pdf2.hash
My Locked PDF2.pdf:password1

1 password hash cracked, 0 left
```

Screenshot 5: John the Ripper cracks My Locked PDF2.pdf – password ‘password1’

Result – My Locked PDF3.pdf: cracked instantly. Recovered password: 1qaz2wsx.

A screenshot of a terminal window titled "MT Linux (Week1-Lab-Configured) [Running] - Oracle VirtualBox". The terminal shows a user at the prompt "(kali㉿kali)-[~/W3-PM1]" running the command "\$ /usr/share/john/pdf2john.pl 'My Locked PDF3.pdf' > hashes/pdf3.hash". Then, they run "\$ john --wordlist=/usr/share/wordlists/rockyou.txt hashes/pdf3.hash". The output shows it loaded 1 password hash, cost 1, and ran 4 OpenMP threads. It successfully cracks the password "1qaz2wsx" with a speed of 9.090g/s. Finally, they run "\$ john --show hashes/pdf3.hash" which displays "My Locked PDF3.pdf:1qaz2wsx" and reports "1 password hash cracked, 0 left".

```
(kali㉿kali)-[~/W3-PM1]
$ /usr/share/john/pdf2john.pl "My Locked PDF3.pdf" > hashes/pdf3.hash

(kali㉿kali)-[~/W3-PM1]
$ john --wordlist=/usr/share/wordlists/rockyou.txt hashes/pdf3.hash
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
1qaz2wsx (My Locked PDF3.pdf)
1g 0:00:00:00 DONE (2026-09-26 07:08) 9.090g/s 13963p/s 13963c/s 13963C/s snuggles..mexico1
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.

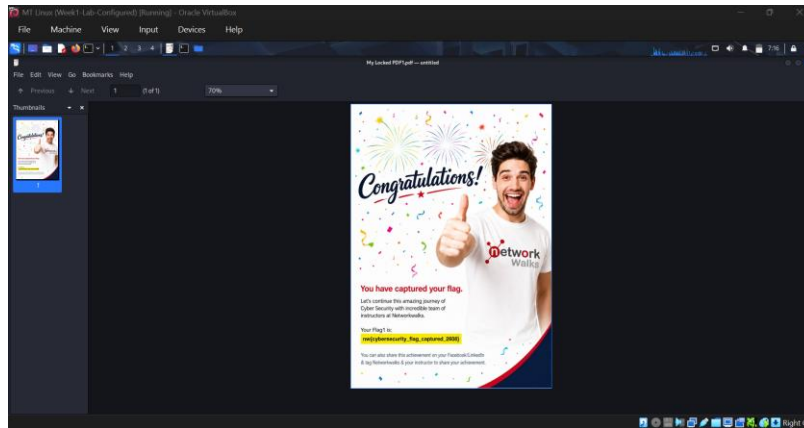
(kali㉿kali)-[~/W3-PM1]
$ john --show hashes/pdf3.hash
My Locked PDF3.pdf:1qaz2wsx

1 password hash cracked, 0 left
```

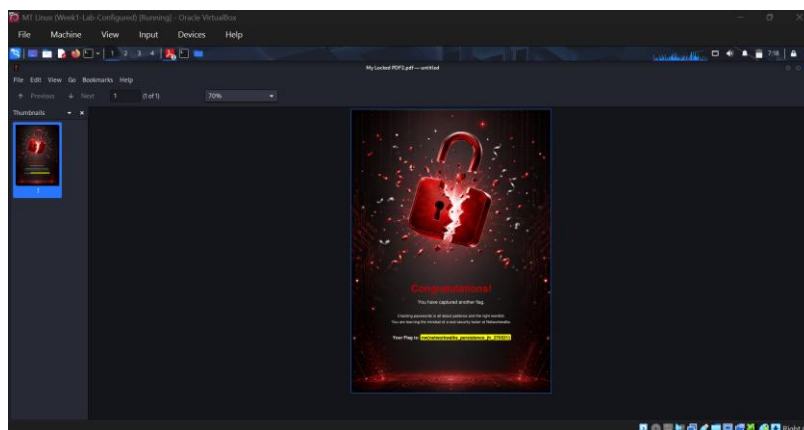
Screenshot 6: John the Ripper cracks My Locked PDF3.pdf – password ‘1qaz2wsx’

Step 4 – Unlocking the PDFs & Capturing Flags

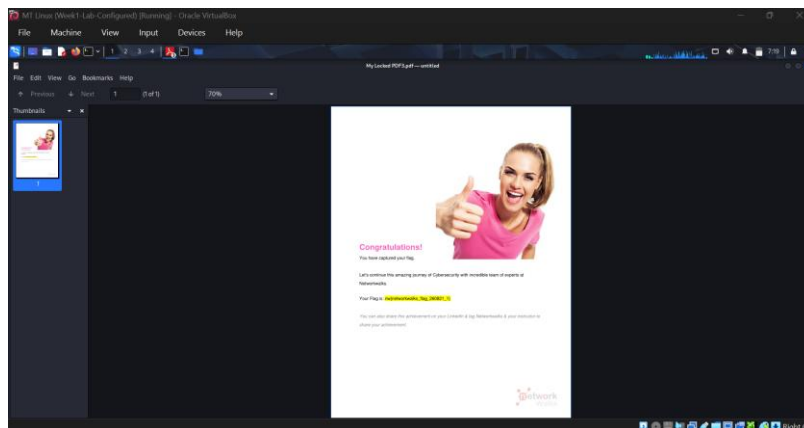
Using the recovered passwords, I opened each PDF in a document viewer to confirm the password worked and captured the embedded verification flag on the unlocked page.



Screenshot 7: My Locked PDF1.pdf unlocked – Flag1: nw{cybersecurity_flag_captured_2608}



Screenshot 8: My Locked PDF2.pdf unlocked – Flag: nw{networkwalks_persistence_jtr_270521}



Screenshot 9: My Locked PDF3.pdf unlocked – Flag: nw{networkwalks_flag_260821_1}

4.2 Online Password Cracking with the Networkwalks Suite (W3-PM2)

As a second, GPU/Linux-independent approach, I repeated the same recovery process using two browser-based Networkwalks tools: the Hash Calculator (to extract the \$pdf\$ hash) and the Password Cracker (to run a dictionary attack against that hash).

Step 1 – Online Hash Extraction (Hash Calculator)



Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF.



TextFilePDF

Click to choose a PDF, or drop one here
Nothing is uploaded; the PDF is parsed locally.

My-Locked-PDF1.pdf | 65.2 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

```
$pd($4*$12B*.1060*1*16*5$diaSc1d175da449753199e44971d32*32*777f4021a7f3c5ae59
8c9c6495c7476e0000000000000000000000000000000*32*ccecdac74b19b5a62688d3b3524e
1374c955cb9cc)c45316494d9446ef81af1
```

CopyDownload

Revision: R4 Version: V4 Key Length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

Free tool by NetworkWalks Academy | Cybersecurity & Ethical Hacking with AI



Hash Calculator

Generate MD5, SHA-1, SHA-256, SHA-384, SHA-512, or extract a crackable hash from a password-protected PDF



Text

File

PDF



Click to choose a PDF, or drop one here

Nothing is uploaded, the PDF is parsed locally

My Locked PDF2.pdf | 204.7 KB

This PDF is encrypted. A crackable hash has been extracted below (pdf2john / hashcat compatible format).

PDF Hash

```
$pdf$4*4*128*-1028*1*16*0853f2cde0ef15b1c0f93ed229d3b1ad*32*8f13ce5aa39ad974364d36a087da76790021446990b9e4114071a4d9104984c1*32*ceecdac74b19b5ac62688d3b3524e1374c955cb9cc3c45316494d9446ef81af1
```

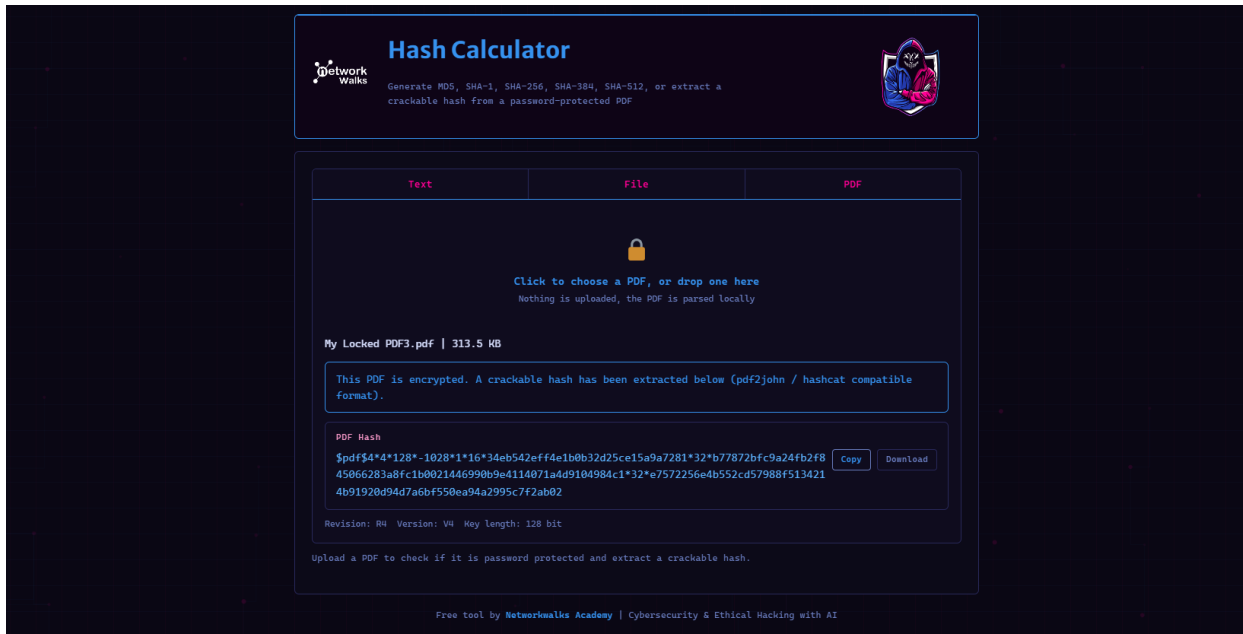
Copy

Download

Revision: RH Version: VH Key Length: 128 bit

Upload a PDF to check if it is password protected and extract a crackable hash.

Free tool by [NetworkWalks Academy](#) | Cybersecurity & Ethical Hacking with AI

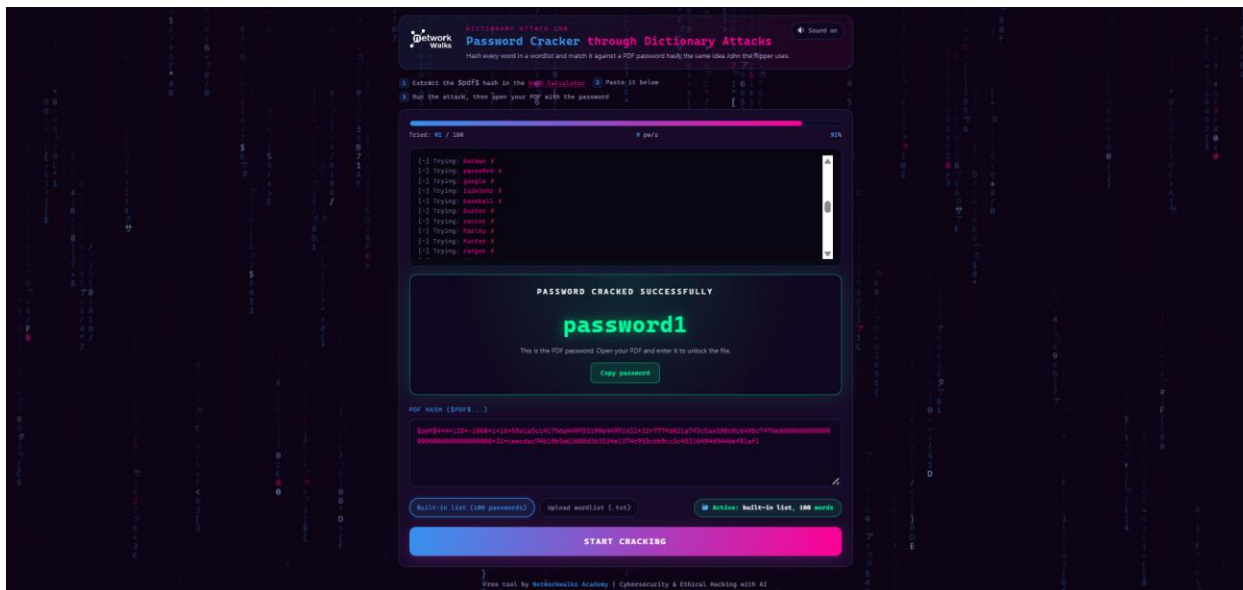


Screenshot 12: \$pdf\$ hash extracted online for My Locked PDF3.pdf

Step 2 – Online Dictionary Attack (Password Cracker)

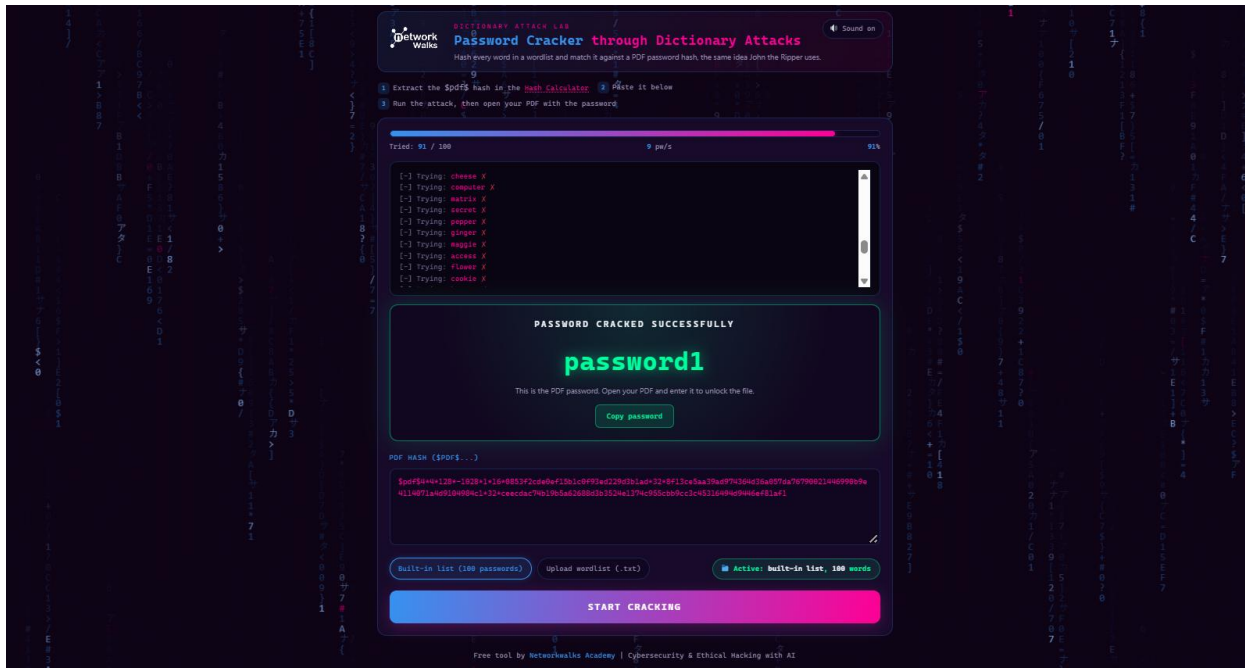
I pasted each extracted hash into the Networkwalks Password Cracker (Dictionary Attack Lab) and ran the built-in 100-word list against it.

Result – PDF1 hash: password cracked successfully – password1 (91/100 words tried).



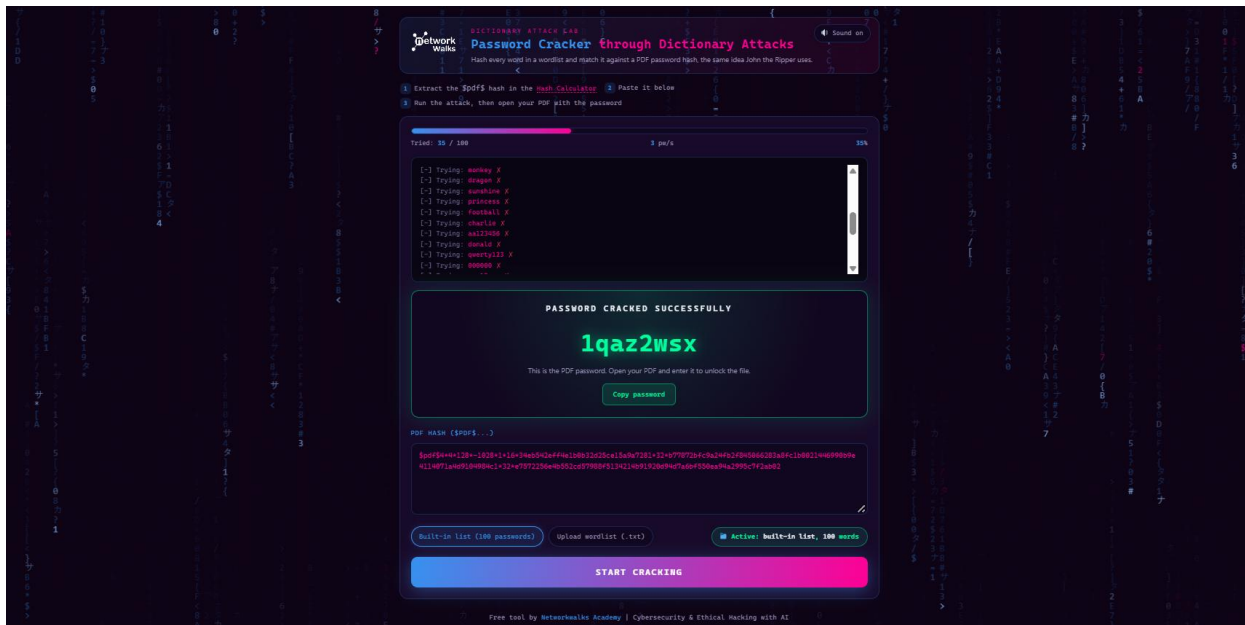
Screenshot 13: Online dictionary attack recovers 'password1' for PDF1

Result – PDF2 hash: password cracked successfully – password1 (91/100 words tried).



Screenshot 14: Online dictionary attack recovers 'password1' for PDF2

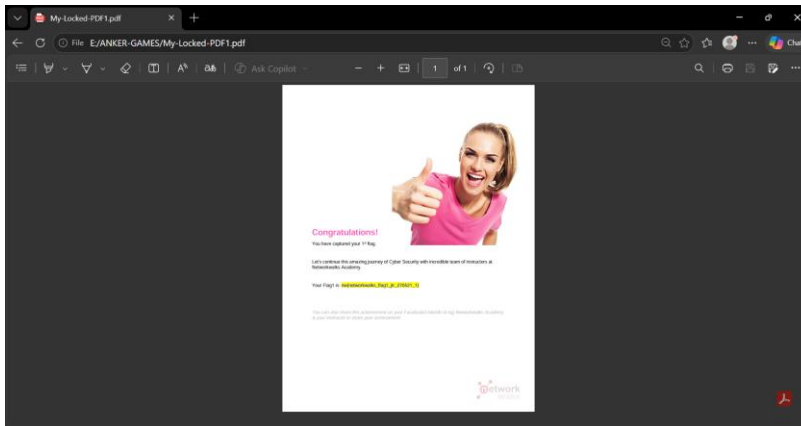
Result – PDF3 hash: password cracked successfully – 1qaz2wsx (35/100 words tried).



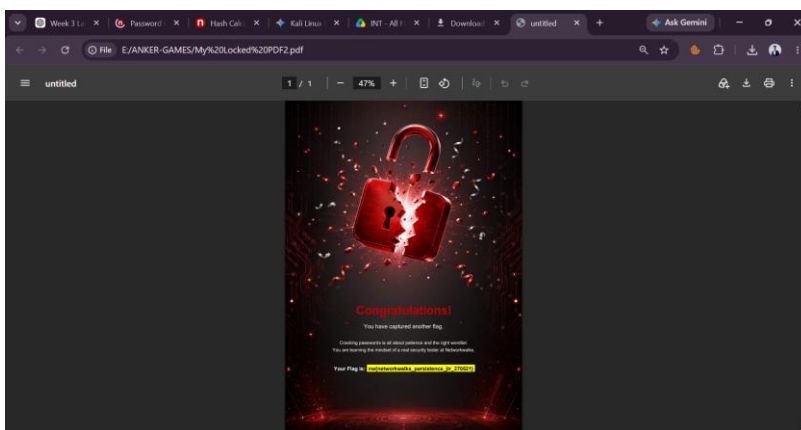
Screenshot 15: Online dictionary attack recovers '1qaz2wsx' for PDF3

Step 3 – Online Verification & Flag Capture

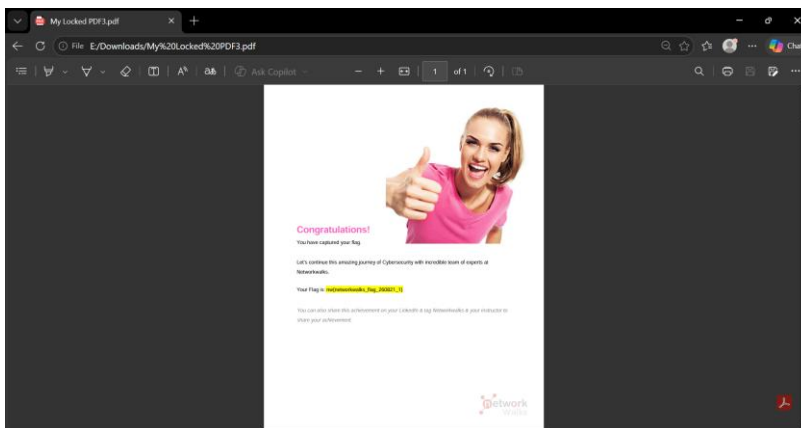
I opened each PDF locally using the password revealed by the online cracker and confirmed the same verification flags were displayed, cross-validating the offline JTR results.



Screenshot 16: PDF1 unlocked online – Flag1: `nw{networkwalks_flag1_jtr_270521_1}`



Screenshot 17: PDF2 unlocked online – Flag: `nw{networkwalks_persistence_jtr_270521}`



Screenshot 18: PDF3 unlocked online – Flag: `nw{networkwalks_flag_260821_1}`

5. Errors, Issues & Root Cause Analysis (Troubleshooting)

During the execution of this lab, several environment and administrative issues were encountered while transferring and processing the target PDFs in Kali Linux. Below is the documentation of these challenges, their root causes, and how they were resolved.

Issue 1 – Missing Source Files inside Kali Linux

Symptom: Running `mv ~/Desktop/*.pdf ~/W3-PM1/` returned ‘No such file or directory’, and `find ~ -iname "*.pdf"` only returned library assets from Python packages.

Root Cause: The target PDF files existed on the Windows Host OS (E:\ANKER-GAMES\week3) and had not yet been mounted or transferred into the Kali Linux virtual machine.

Correction / Solution: Mounted the VirtualBox Shared Folder to `/mnt/week3` using `sudo mount -t vboxsf week3 /mnt/week3`.

Issue 2 – Empty Mount Directory

Symptom: `ls -la /mnt/week3/` listed total 0 files, and the `cp` command failed with nothing to copy.

Root Cause: The shared folder settings in VirtualBox pointed to an incorrect or unrefreshed Windows folder path.

Correction / Solution: Re-configured the VirtualBox Shared Folder with Auto-mount and Make Permanent enabled, then re-mounted the directory.

Issue 3 – Permission Denied during File Copy

Symptom: Executing `cp /mnt/week3/*.pdf ~/W3-PM1/` as user `kali` returned ‘Permission denied’.

Root Cause: The `~/W3-PM1` directory had previously been created using `sudo mkdir -p`, assigning ownership strictly to `root:root`.

Correction / Solution: Transferred directory ownership back to the standard user using `sudo chown -R kali:kali ~/W3-PM1`.

Issue 4 – Missing PDF Viewer Dependency

Symptom: Attempting to view the cracked PDFs with `evince` “My Locked PDF1.pdf” failed because the GNOME Document Viewer was not installed by default.

Root Cause: Kali Linux ships `atril` / `qpdfview` as its default lightweight PDF viewers instead of `evince`.

Correction / Solution: Opened the files directly using the default system viewer / XFCE File Manager GUI instead of `evince`.

6. Summary of Recovered Credentials & Flags

The table below consolidates the passwords and flags recovered through both the offline (JTR) and online (Networkwalks) methods for cross-verification.

Document Name	Method Used	Recovered Password	Captured Flag
My Locked PDF1.pdf	JTR (offline)	good-luck	nw{cybersecurity_flag_captured_2608}
My Locked PDF1.pdf	Networkwalks (online)	password1	nw{networkwalks_flag1_jtr_270521_1}

Document Name	Method Used	Recovered Password	Captured Flag
My Locked PDF2.pdf	JTR (offline)	password1	nw{networkwalks_persistence_jtr_270521}
My Locked PDF2.pdf	Networkwalks (online)	password1	nw{networkwalks_persistence_jtr_270521}
My Locked PDF3.pdf	JTR (offline)	1qaz2wsx	nw{networkwalks_flag_260821_1}
My Locked PDF3.pdf	Networkwalks (online)	1qaz2wsx	nw{networkwalks_flag_260821_1}

7. Risk Analysis / Impact

Based on the information collected during this password-recovery exercise, the following potential risks were identified.

Risk level key: ● Critical (Red) ● Medium (Orange) ● Low (Green)

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Weak, dictionary-guessable PDF passwords	All three PDFs (good-luck, password1, 1qaz2wsx) were cracked within seconds using the default rockyou.txt wordlist	Confidential document contents can be exposed almost instantly to any attacker with a copy of the file	● Critical
2	PDF encryption alone is not sufficient protection	Both offline (JTR) and online (Networkwalks) tools extracted a crackable \$pdf\$ hash directly from the file metadata without needing the password first	Encryption gives a false sense of security if the underlying password is weak	● Medium
3	Free online hash/password tools can process sensitive files	The Networkwalks Hash Calculator and Password Cracker accepted the PDFs and produced usable \$pdf\$ hashes in the browser	Uploading real confidential PDFs to unknown third-party web tools could leak sensitive data outside the organisation	● Medium
4	Local environment configuration errors during setup	Missing shared folder mounts, root-owned directories and a missing PDF viewer package delayed the exercise (see Section 5)	Misconfigured lab/production environments can slow incident response or, in a real assessment, cause evidence-handling mistakes	● Low

8. Recommendations

Based on the findings, the following security measures are recommended:

- Enforce strong PDF passwords – require long, random, non-dictionary passphrases (12+ characters) for any confidential PDF instead of common words or keyboard patterns like 1qaz2wsx.
- Never rely on PDF password protection alone – combine it with proper access control, encryption at rest, and restricted sharing rather than treating a password as the only safeguard.
- Avoid uploading sensitive real documents to third-party online tools – use offline tools such as John the Ripper inside an isolated lab environment when the file is genuinely confidential.

- Rotate and audit shared lab/demo credentials – since these sample passwords are publicly guessable, they should never be reused for real organisational documents.
- Fix environment configuration before an assessment – pre-verify shared folders, directory ownership, and required packages (pdf2john.pl, PDF viewers) to avoid delays during time-boxed engagements.
- Maintain a clear evidence trail – keep hash files, cracked passwords, and screenshots (as done in this report) for every recovery activity to support audit and review.
- Perform password-recovery testing only with authorisation – all cracking activity here was carried out strictly on self-owned / internship-provided lab files.

9. Conclusion

During Week 3 of my Cybersecurity & Ethical Hacking internship at Networkwalks, I completed practical activities covering PDF password recovery and hash cracking through both offline and online methods.

In the offline activity (W3-PM1), I used pdf2john.pl to extract crackable \$pdf\$ hashes inside Kali Linux and used John the Ripper with the rockyou.txt wordlist to recover all three passwords - good-luck, password1 and 1qaz2wsx - within seconds. I also resolved several real-world environment issues around shared folder mounting, file permissions and missing PDF viewer packages.

In the online activity (W3-PM2), I used the Networkwalks Hash Calculator and Password Cracker to independently extract and crack the same category of \$pdf\$ hashes from a browser, without needing a Linux environment or local GPU, and cross-verified the results by unlocking each PDF and capturing its embedded flag.

These exercises demonstrated that PDF password protection is only as strong as the password chosen - dictionary-based passwords, however seemingly ‘random’ (like 1qaz2wsx), are cracked almost instantly with a common wordlist. I also learned that both a fully offline, script-driven workflow and a simplified web-based workflow can achieve the same result, giving flexibility depending on the environment available.

Finally, I reinforced the importance of ethics in cybersecurity: all password-recovery activity in this report was performed strictly on self-owned / internship-provided lab files, and the same techniques must never be used against documents without proper authorisation.

— End of Report —

 **Author**

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★ Program: Cybersecurity at Networkwalks | Week: 03